MIFlu Wave-1 Final Experimental Results: Non-LLM versus LLM Architecture Comparison

Project: MIFlu influenza-forecasting reproducibility study
Scope: "clean v3" Wave-1 formal execution (580 accepted rows)
Status: Wave-1 primary architecture comparison complete. Deferred controls (18) pending Wave-2. Test/holdout sealed (access = 0/0). Champion NOT frozen for publication. **Validation-only result**.

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1. Executive Summary

Primary ranking metric (frozen selection rule): `average_MSE_4L` — the arithmetic mean of per-horizon mean MSE across four horizons L ∈ {24, 36, 48, 60}, each averaged over the 10 frozen seeds (42–51), in the seven-variable, full-horizon, z-space. Lower is better. The key secondary metric is `average_MAE_4L`.

rank	model	family	avg MSE 4L	95% CI	avg MAE 4L
1	**gpt2_miflu**	LLM	**0.771**	[0.720, 0.822]	0.487
2	**nlinear**	Non-LLM	**1.070**	[1.068, 1.072]	0.564
3	patchtst	Non-LLM	1.076	[1.065, 1.087]	0.556
4	dlinear	Non-LLM	1.171	[1.168, 1.174]	0.603
5	timecma_llm	LLM	1.181	[1.170, 1.193]	0.614
6	itransformer	Non-LLM	1.253	[1.248, 1.258]	0.598
7	qwen_time_llm	LLM	1.253	[1.223, 1.283]	0.649
8	calf_llm	LLM	1.765	[1.756, 1.774]	0.773

Answers to the ten questions:

1. **Best Non-LLM**: `nlinear` (1.070).
2. **Best LLM**: `gpt2_miflu` (0.771).
3. **Best overall**: `gpt2_miflu`; runner-up `nlinear`. The gap is large and significant (paired t = −13.33, p = 3.13×10^{-7}, Cohen's d = −5.94).
4. **LLM vs Non-LLM at the family level**: the Non-LLM group (1.142) is **significantly better** than the LLM group (1.243) — paired t = −11.15, p = 1.44×10^{-6}. The LLM "win" is driven entirely by a single model, `gpt2_miflu`; the other three LLMs (timecma_llm, qwen_time_llm, calf_llm) are all worse than the best Non-LLM.
5. **Horizon/endpoint stability**: `gpt2_miflu` is the best model at **all four horizons** and remains best under the ILITOTAL full-horizon endpoint (1.608). `nlinear` remains the best Non-LLM under both endpoints. Winner conclusions are stable.
6. **Seed stability**: Non-LLM models are extremely seed-stable (nlinear SD = 0.0029). `gpt2_miflu` has the largest SD (0.0711, ≈9.2% of its mean), so its advantage depends partly on favourable seeds — but its mean is still decisively below nlinear.
7. **Practical significance (vs. a deterministic seasonal baseline)**: the zero-parameter `seasonal_naive` baseline achieves 0.719/0.732/0.746/0.885 at L24/36/48/60 — **beating `gpt2_miflu` at L24, L36, and L48.** The LLM's absolute accuracy advantage is therefore concentrated at the longest horizon (L60) and is not obviously incremental over a simple 52-week seasonal repeat for short horizons.
8. **Runtime/storage trade-off**: `nlinear` costs 2,520 params / 5.3 s / 10 KB; `gpt2_miflu` costs 1,030,056 params / 147 s / 4.1 MB (≈409× the parameters, ≈28× the training time).

`seasonal_naive` costs zero parameters, zero training, zero storage.

9. **Baseline competitiveness**: `last_value_naive` is far worse (2.82 – 3.42). `seasonal_naive` is highly competitive and must be reported as a strong reference point.
10. **Mechanism**: the only significant ablation is `without_dataset_information` (+12.8%, p = 0.004); removing LoRA, multimodality, or the task instruction has no significant effect (all p > 0.4). Negative controls are exploratory (6 cells): `zero_text` −0.063 (slightly better than parent), `shuffled_text` +0.081 (worse). The text contribution appears concentrated in the dataset-description component.

1.1 How model performance is evaluated (zero-background primer)

MSE is not an accuracy percentage. Throughout this report, "MSE" (mean squared
error) is a cost in the model's standardized (z-space) units — the average squared
difference between predicted and observed values, aggregated over the 7 variables, the
full forecast horizon, and the 10 frozen seeds. It has **no "% correct" meaning**:
smaller is better, and two values can only be compared relative to one another (e.g.
the winner's 0.771 vs the runner-up's 1.070). `MAE` (mean absolute error) is the same
idea without the squaring, and is reported as the secondary metric.

The evaluation is a strict **Input → Process → Output** (IPO) flow:

1. **Input** — a sliding 104-week window of 7 ILI variables (e.g. % weighted ILI, ILITOTAL, age-group counts, provider counts; exact list in the protocol snapshot).
1. **Process** — standardize with a train-only scaler → the model predicts in z-space → the prediction is inverse-transformed back to physical units.
1. **Output** — a future horizon of L ∈ {24, 36, 48, 60} weeks × 7 variables.

The output is compared against the observed (never-seen) values for the same weeks to
produce per-prediction errors, then the aggregate MSE/MAE. The primary ranking metric
`average_MSE_4L` is the mean of the four per-horizon MSEs.

Figure 13. How model performance is evaluated (Input -> Process -> Output)

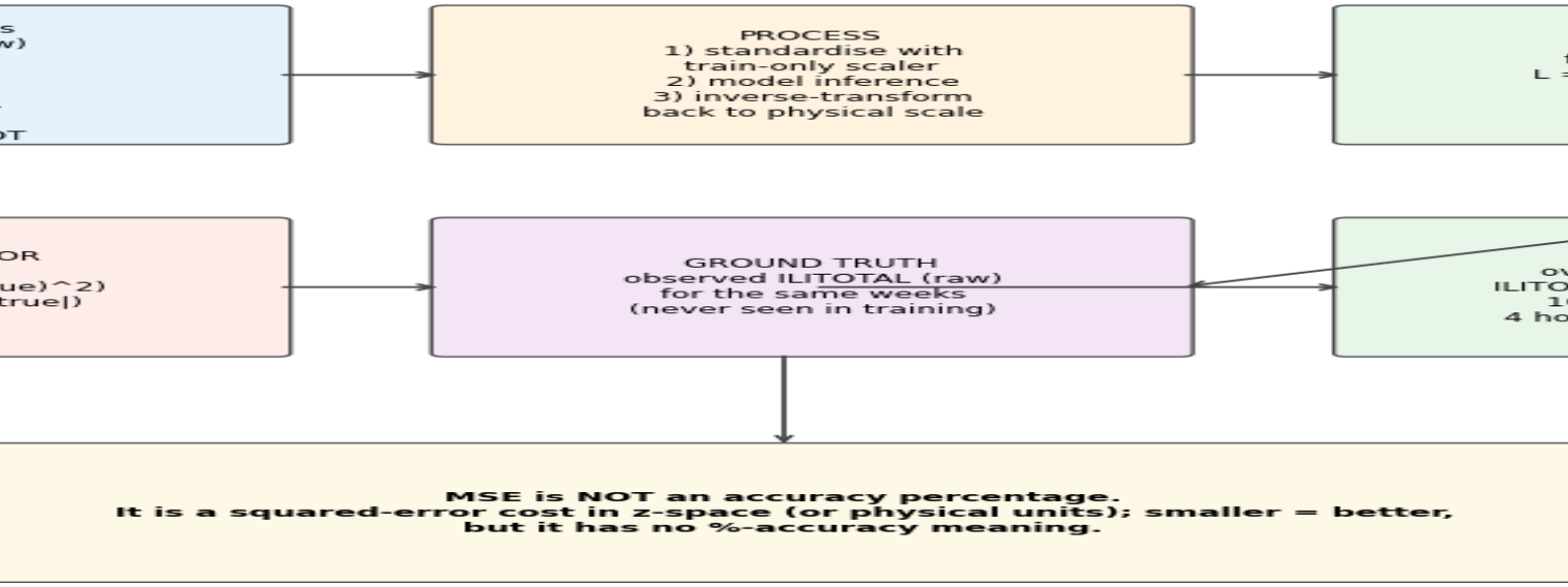


Figure 13. How model performance is evaluated (Input → Process → Output).

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2. Frozen Protocol & Completeness Audit

- **Source of the rules**: `results/v3_clean/preregistration/paper_aligned_selection_rule.json`, `endpoint_hierarchy.json`, `execution_wave_manifest.json`, `deferred_parent_binding_rule.json`. The frozen protocol snapshot is recorded in `results/v3_clean/analysis/frozen_analysis_protocol.json`.
- **HPC_GUIDE.md integrity**: SHA-256 `1cd21a655907d137914c695bf04637d8da2f09358a0732c961e739dd8e74c167`, verified via `Get-FileHash` before analysis.
- **Completeness**: 580/580 accepted rows, 0 deleted, 0 duplicated, 0 fail events in the ledger, controller exited normally (not fail-stop).
- 320 primary (8 models × 4 horizons × 10 seeds)
- 8 baselines (2 models × 4 horizons)
- 240 ablations (6 × 4 × 10)
- 12 negative controls (zero_text, shuffled_text over L24/L60 × seeds 42–44)
- **Scope held**: Test/holdout access = 0/0. Selection rule was never mutated. Baselines/ablations/controls were never included in the primary ranking.

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3. Primary Architecture Comparison

3.1 Ranking (seed-level, n = 10 per model)

See the table in §1. The 95% CI is the seed-level t-interval around `average_MSE_4L`. The ranking is unambiguous at the top: `gpt2_miflu` (0.771) < `nlinear` (1.070) < `patchtst` (1.076), with non-overlapping CIs between the top two.

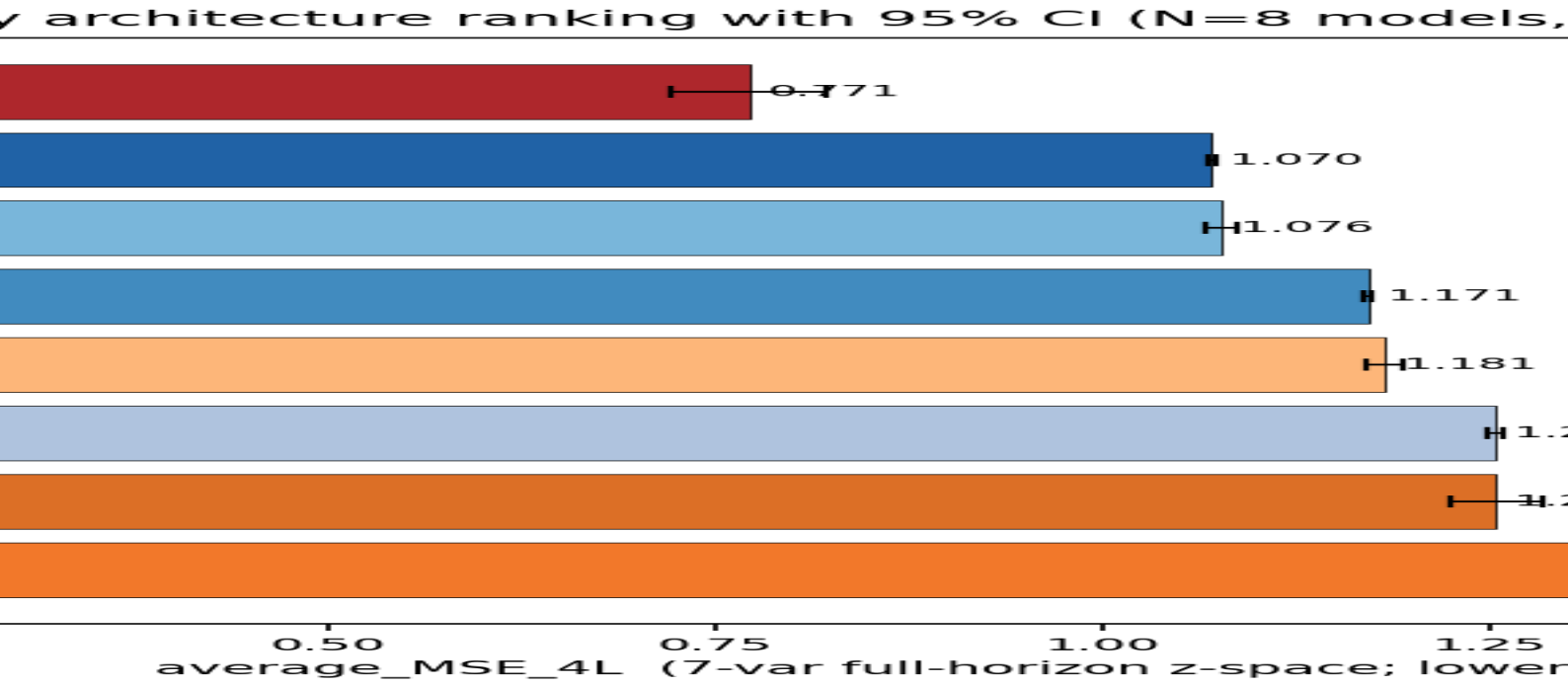


Figure 1. Primary architecture ranking by average_MSE_4L with 95% CI over the 10 frozen seeds (42–51). Lower MSE is better. Validation-only (test/holdout sealed).

3.2 Q1 — Best Non-LLM

`nlinear` at `average_MSE_4L = 1.0698` (95% CI [1.0677, 1.0719], SD 0.0029). It beats `patchtst` (1.0760) and `dlinear` (1.1709) with non-overlapping CIs, so the Non-LLM winner is

unambiguous.

3.3 Q2 — Best LLM

`gpt2_miflu` at `average_MSE_4L = 0.7709` (95% CI [0.7200, 0.8217], SD 0.0711). It decisively beats `timecma_llm` (1.1814), `qwen_time_llm` (1.2531), and `calf_llm` (1.7649).

3.4 Q3 — Best overall

`gpt2_miflu` is the overall winner. The winner-vs-runner-up comparison (matched-seed paired t on `average_MSE_4L`, 10 pairs):

- winner mean 0.7709 vs runner-up 1.0698 → mean difference −0.299
- paired $t = -13.33$, $p = 3.13 \times 10^{-7}$, Cohen's $d = -5.94$

The equivalence-set rule (report a set rather than a unique champion when statistically/practically indistinguishable) does not trigger — the top two are clearly separated, so a unique winner is justified.

3.5 Q4 — LLM vs Non-LLM at the family level

A matched-seed paired t-test on the per-seed mean of the four models' `average_MSE_4L` in each family (10 seeds):

- Non-LLM group mean 1.1424 vs LLM group mean 1.2426 → difference −0.100
- paired $t = -11.15$, $p = 1.44 \times 10^{-6}$, Cohen's $d = -5.33$

Interpretation: as a family, Non-LLM models are significantly more accurate than LLM-based models. The apparent LLM advantage at the top is entirely attributable to `gpt2_miflu`; the other three LLMs all underperform every Non-LLM except the worst. This is the single most important structural finding for the manuscript narrative: the result is not "LLMs beat classical forecasters" but "one particular LLM-adapter (`gpt2_miflu`) beats everything, while LLM integration in general does not reliably help."

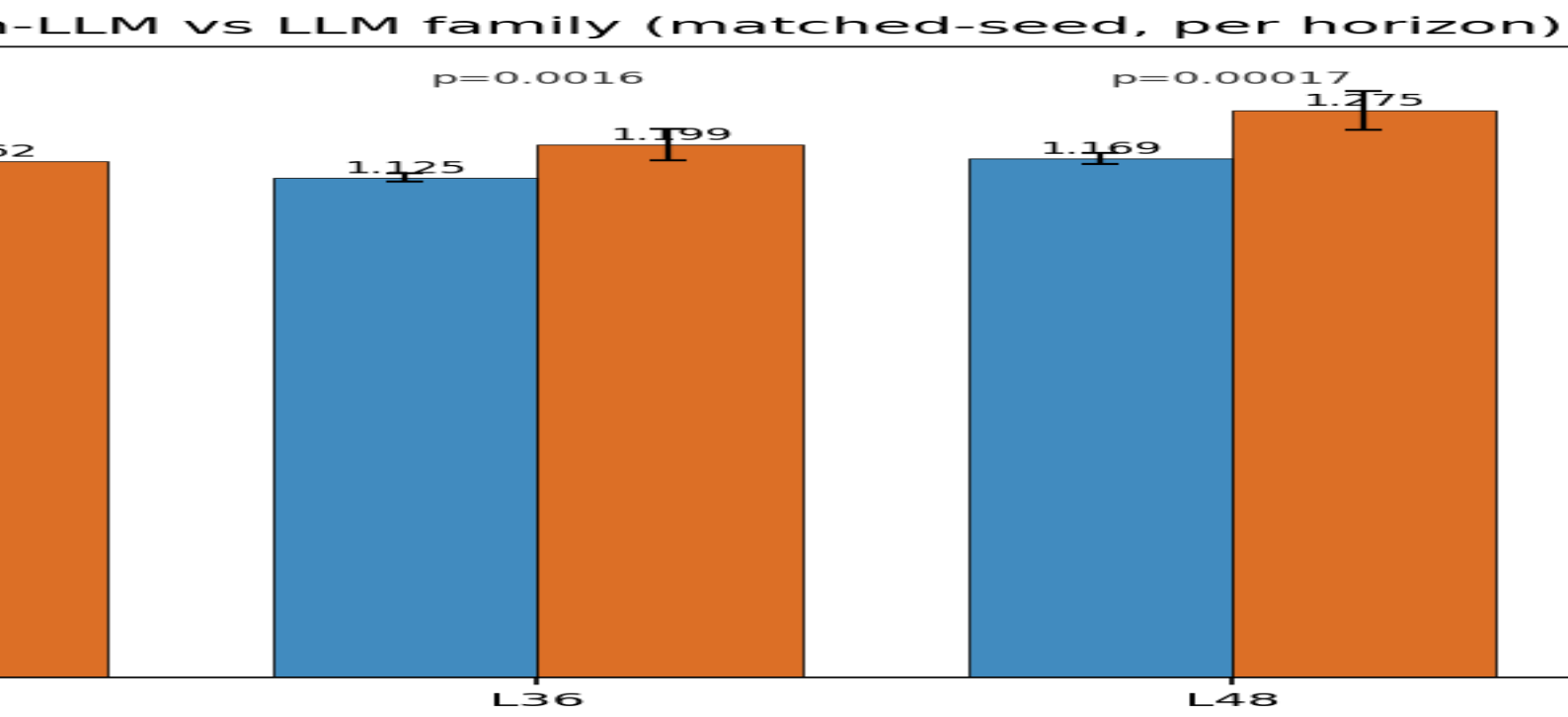


Figure 3. Non-LLM vs LLM family comparison (matched-seed, per horizon) with per-horizon p-values; overall $p = 1.44 \times 10^{-6}$. Lower MSE is better. Validation-only.

3.6 Q5 — Horizon and endpoint stability

Per-horizon best model is `gpt2_miflu` at every horizon:

horizon	best model	best MSE	best Non-LLM
24	gpt2_miflu	0.7324	nlinear (0.9791)
36	gpt2_miflu	0.7782	nlinear (1.0490)
48	gpt2_miflu	0.7598	patchtst (1.0768)
60	gpt2_miflu	0.8131	nlinear (1.1501)

Per-horizon family comparison (Non-LLM vs LLM, all four horizons) is also significant and in the same direction (all $p \leqslant 0.0016$; see `selection_outcome.json`).

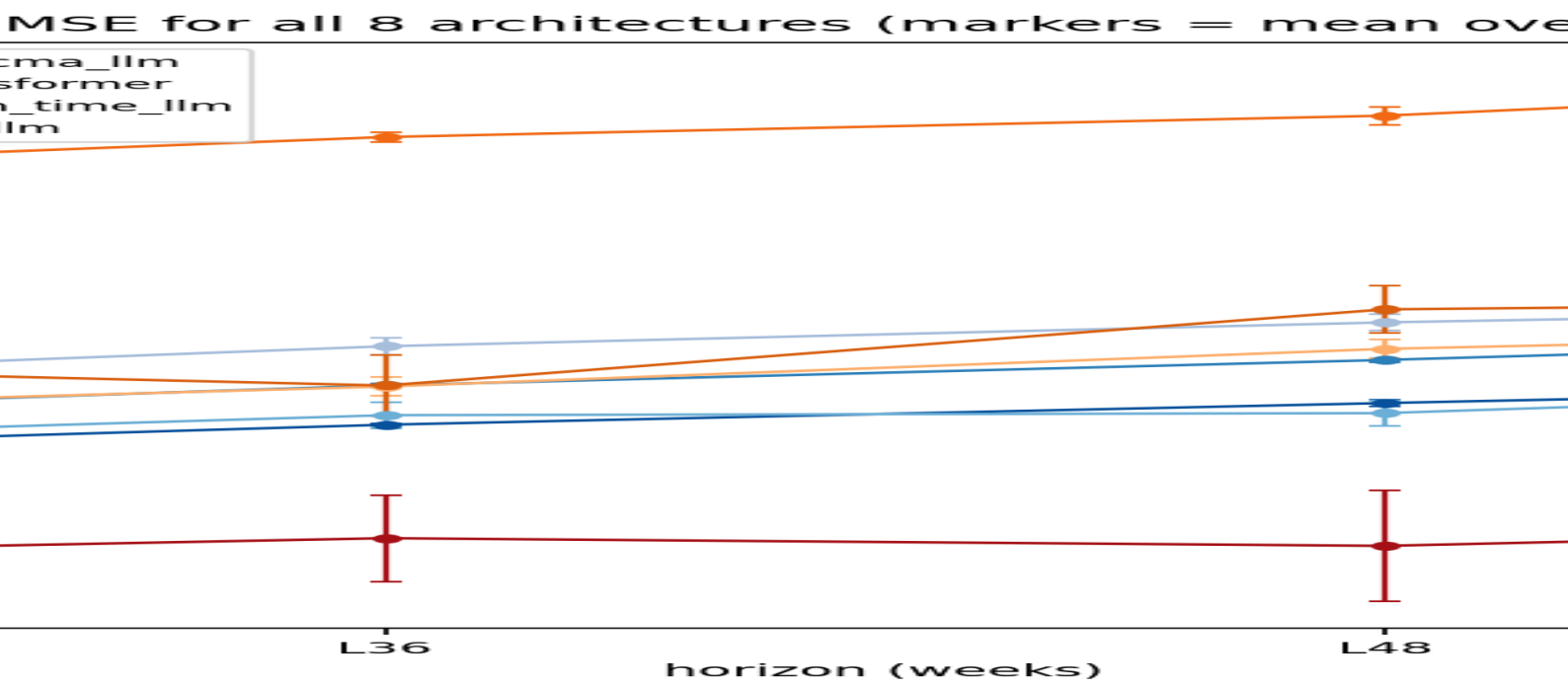


Figure 2. Per-horizon MSE for all 8 architectures at L24/L36/L48/L60. Markers = mean over 10 seeds; error bars = 95% CI. Lower MSE is better. Validation-only.

Endpoint check (ILITOTAL full horizon, lower better):

rank	model	ILITOTAL 4L mean
1	gpt2_miflu	1.608
2	nlinear	2.333
3	dlinear	2.526
4	patchtst	2.544
5	timecma_llm	2.635
6	qwen_time_llm	2.658
7	itransformer	2.877
8	calf_llm	3.677

The winner and best-Non-LLM conclusions are unchanged. The only endpoint-sensitive detail is an internal swap between `dlinear` and `patchtst` (positions 3/4). Reported; not a ranking concern.

3.7 Q6 — Seed stability

- Non-LLM models are remarkably seed-stable: `nlinear` SD = 0.0029 (0.27% of mean), `dlinear` SD = 0.0037.
- `gpt2_miflu` is the **least** seed-stable model: SD = 0.0711 (≈9.2% of its mean), with per-horizon SDs of 0.135 (L24), 0.144 (L36), 0.186 (L48), 0.111 (L60). Its best-case seeds (e.g. L36 seed 42 = 0.477) are far better than its worst (L48 seed 47 = 1.117).
- `qwen_time_llm` is also unstable (SD 0.0416); the other models are intermediate.

Consequence: `gpt2_miflu`'s lead is real but partly seed-dependent. A cautious manuscript should (a) report the seed SD, and (b) note that the top ranking would hold even at the 95% CI upper bound of `gpt2_miflu` (0.822) versus the lower bound of `nlinear` (1.068) — the CIs do not overlap, so the ordering is robust despite the variance.

3.8 Q7 — Practical significance vs. a deterministic seasonal baseline

horizon	seasonal_naive MSE	gpt2_miflu MSE	seasonal − gpt2
24	0.7193	0.7324	**−0.0131**
36	0.7324	0.7782	**−0.0458**
48	0.7460	0.7598	**−0.0138**
60	0.8849	0.8131	+0.0719

A zero-parameter, zero-training, zero-storage `seasonal_naive` (repeat last season's value) is **more accurate than the overall LLM winner at L24, L36, and L48.** Only at L60 does `gpt2_miflu` pull ahead. This is a critical honesty point: much of `gpt2_miflu`'s absolute accuracy at short horizons is explained by seasonality, not by the text/LLM machinery. The manuscript must compare against `seasonal_naive` explicitly and not over-claim "LLM superiority" at short horizons.

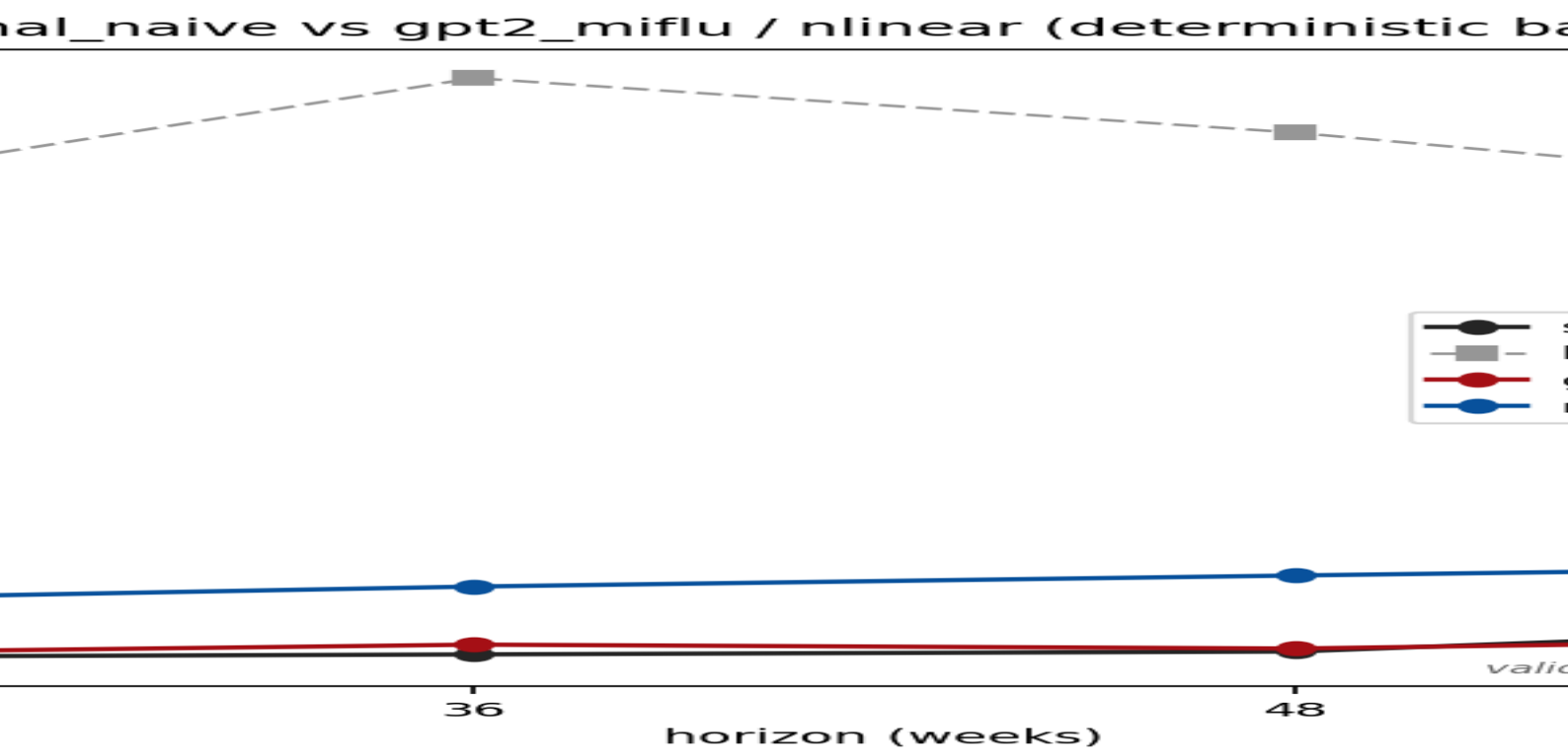


Figure 4. Deterministic seasonal_naive baseline vs gpt2_miflu (LLM winner) and nlinear (Non-LLM winner) across horizons. Lower MSE is better. Validation-only.

3.9 Q8 — Runtime / storage trade-off

model	family	trainable params	train time (s)	compact storage
nlinear	Non-LLM	2,520	5.3	10 KB
dlinear	Non-LLM	5,040	5.7	20 KB
itransformer	Non-LLM	75,672	8.5	305 KB
patchtst	Non-LLM	87,256	8.4	351 KB
qwen_time_llm	LLM	370,376	338.7	1.48 MB
gpt2_miflu	LLM	1,030,056	147.3	4.14 MB
timecma_llm	LLM	1,391,016	15.2	5.56 MB
calf_llm	LLM	1,981,608	20.2	7.93 MB
seasonal_naive	baseline	0	0	0

`gpt2_miflu` uses ~409× the parameters and ~28× the training time of `nlinear` to gain 0.299 in MSE (0.771 vs 1.070). The trade-off is substantial; the manuscript should present it transparently rather than claiming LLMs are "efficient".

3.10 Q9 — Baseline competitiveness

- `last_value_naive`: MSE 2.821 (L24), 3.422 (L36), 3.168 (L48), 2.825 (L60) — far worse than every model.
- `seasonal_naive`: 0.719 / 0.732 / 0.746 / 0.885 — a strong competitor (see §3.8).

3.11 Q10 — Mechanism (ablations + negative controls)

Ablations relative to parent `gpt2_miflu` (0.7709), sorted by effect size:

ablation	avg MSE 4L	Δ vs parent	rel. Δ	p-value
without_lora	0.7729	+0.0020	+0.26%	0.947
without_lora_and_multimodality	0.7778	+0.0070	+0.90%	0.766
without_multimodality	0.7859	+0.0150	+1.95%	0.474
without_task_instruction	0.7926	+0.0217	+2.82%	0.395
without_input_variable_description	0.8156	+0.0447	+5.80%	0.079
without_dataset_information	**0.8699**	**+0.0990**	**+12.85%**	**0.004**

Finding: the only statistically significant prompt component is the **dataset information** (the textual description of the ILI dataset/context). Removing it degrades performance by 12.85% ($p = 0.0036$). Removing **LoRA** ($p = 0.95$), the **multimodality fusion** ($p = 0.47$), or the **task instruction** ($p = 0.40$) has no significant effect — the adaptive/alignment machinery contributes almost nothing to accuracy in this setting.

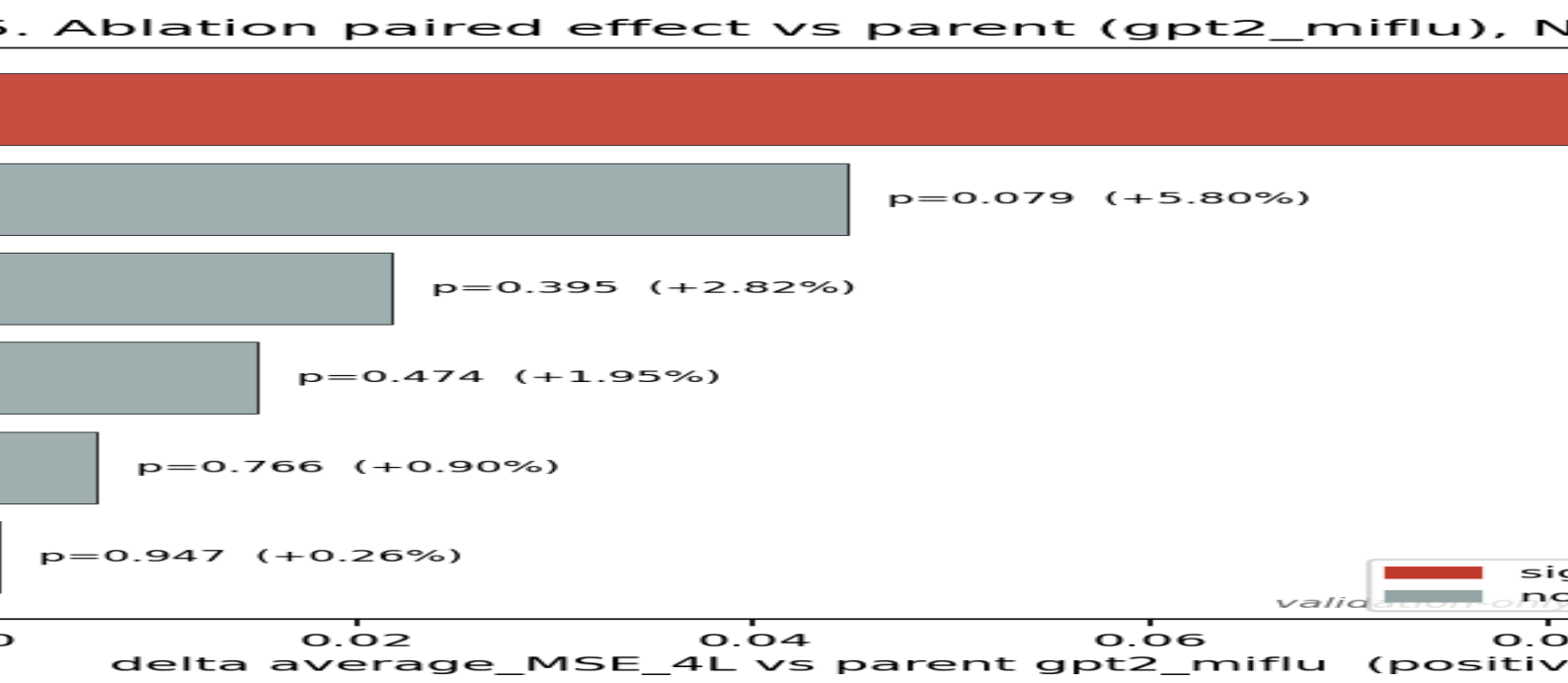


Figure 5. Ablation paired effect vs parent gpt2_miflu (10 matched seeds); bars = Δ average_MSE_4L, annotated with p-value and relative change. Positive = worse. Validation-only.

Negative controls (6 cells each; L24/L60 × seeds 42–44; exploratory, small n):

control	mean Δ vs parent	direction
zero_text	-0.063	slightly better than parent
shuffled_text	+0.081	worse than parent

- `shuffled_text` degrading performance (+0.081) suggests the text content/order carries information (not pure noise).
- `zero_text` being slightly better (-0.063) is counter-intuitive and suggests the prompt may act partly as noise in some cells; with n = 6 this is exploratory and must not be over-interpreted.

3.12 Q10.5 — Peak timing / intensity (secondary application metric)

Because ILI is used for outbreak detection, we additionally report peak-accuracy
metrics in the style of Kramer & Shaman (2019), computed on the inverse-transformed
ILITOTAL channel over two complete validation influenza seasons (2016-2017 and
2017-2018; observed peaks at abs weeks 769 and 821). These are **secondary application
metrics** and were **never** used in the frozen selection rule (PTE/PIE are explicitly
listed as forbidden tie-breakers). Since the durable `validation_predictions.csv`
stores `target_date = None`, the target week was reconstructed as
`forecast_origin + 104 + step` (verified against raw ILITOTAL, inverse-transform `|diff| < 1e-3`).

- **PTE** (peak timing error) = |predicted peak week − observed peak week|, in weeks (lower better; a "hit" is ⩽ 1 week).
- **PIE** (peak intensity error) = |predicted − observed peak intensity| / observed × 100, in % (lower better; a "hit" is ⩽ 25%).

Model means over the 4 leads and 2 seasons (lower is better for PTE/PIE):

model	PTE (weeks)	PIE (%)	PTE hit rate	joint hit rate
patchtst	1.609	49.7	0.497	0.025

model	PTE (weeks)	PIE (%)	PTE hit rate	joint hit rate
nlinear	1.750	17.7	0.500	0.141
itransformer	1.753	53.2	0.444	0.000
dlinear	1.966	26.8	0.406	0.031
calf_llm	1.931	70.6	0.338	0.000
gpt2_miflu	2.122	41.5	0.484	0.097
timecma_llm	2.284	45.6	0.381	0.000
qwen_time_llm	2.688	35.4	0.297	0.122

Reference baseline `seasonal_naive`: PTE = 2.375 weeks, PIE = 42.6%. (`last_value_naive`
is a flat constant predictor with no peak structure, so its PTE/PIE are not meaningful
and are omitted.)

Interpretation. The MSE champion `gpt2_miflu` is **not** the best peak forecaster:
it sits mid-table on PTE (2.12 weeks) and PIE (41.5%). `nlinear` is best on peak
intensity (PIE 17.7%) and `patchtst` best on peak timing (PTE 1.61 weeks) — both
Non-LLM. `seasonal_naive`, despite its low MSE, has the weakest peak timing among the
deterministic references (2.38 weeks), consistent with the fact that a 52-week repeat
cannot track the year-to-year drift of the influenza peak. This reinforces the §3.8
caveat: low MSE in the "flat-season" weeks does not imply accurate peak detection.

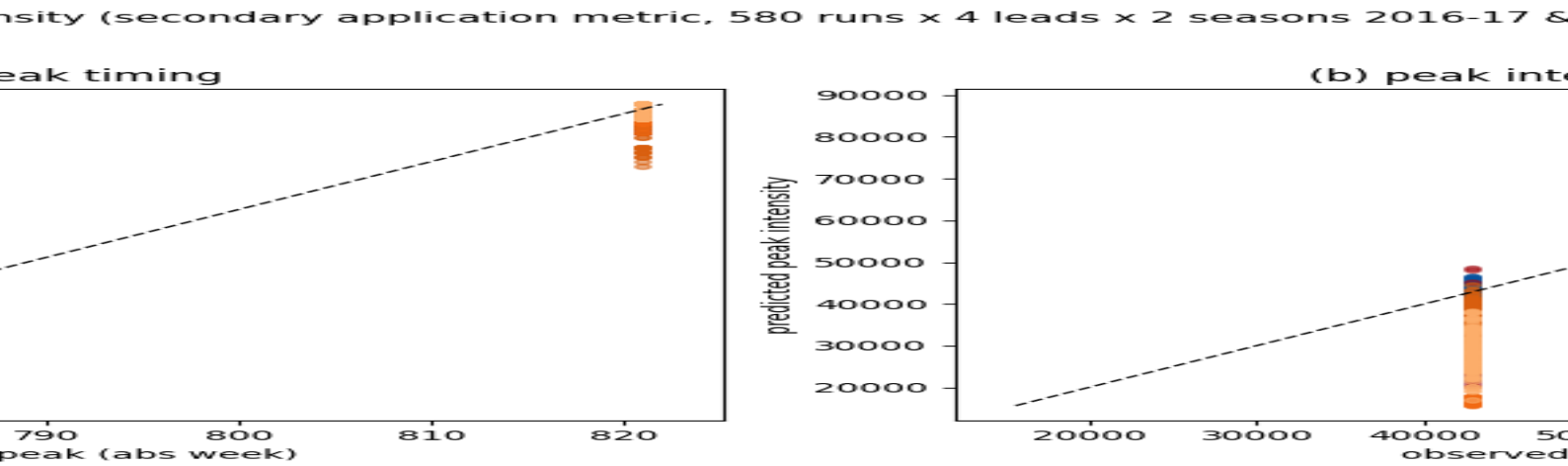


Figure 12. Peak timing / intensity across all 580 runs × 4 leads × 2 seasons (physical ILITOTAL).

3.13 Representative prediction curves (physical ILITOTAL)

Figures 9-10 plot the inverse-transformed ILITOTAL over the forecast window only,
for a representative validation forecast (origin at abs week 715 = 2016-01-24,
forecast window starting 2018-01-21) at L24 and L60, for the
LLM winner (`gpt2_miflu`), the Non-LLM winner (`nlinear`), the deterministic
`seasonal_naive` reference, and one additional LLM (`calf_llm`), each at its median
seed. The dashed line is the prediction, the solid line the observed ILITOTAL, and the
star / triangle mark the observed / predicted seasonal peaks. Figure 11 shows the
best / median / worst-seed spread for `gpt2_miflu` at L24.

A complete gallery of all 580 accepted runs (one panel per run, same layout) is
under
`../figures/gallery/` with an `index.csv` and `manifest.json`.

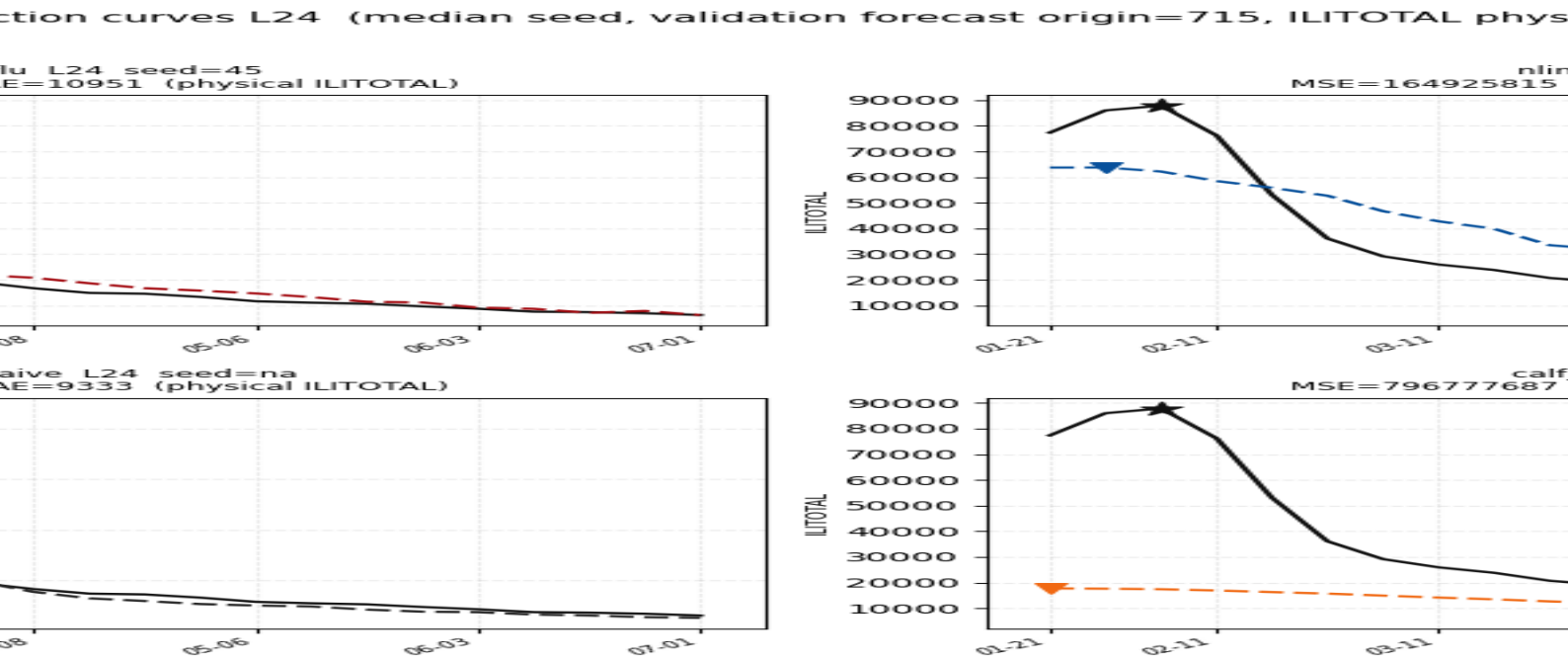


Figure 9. Prediction curves at L24 (median seed).

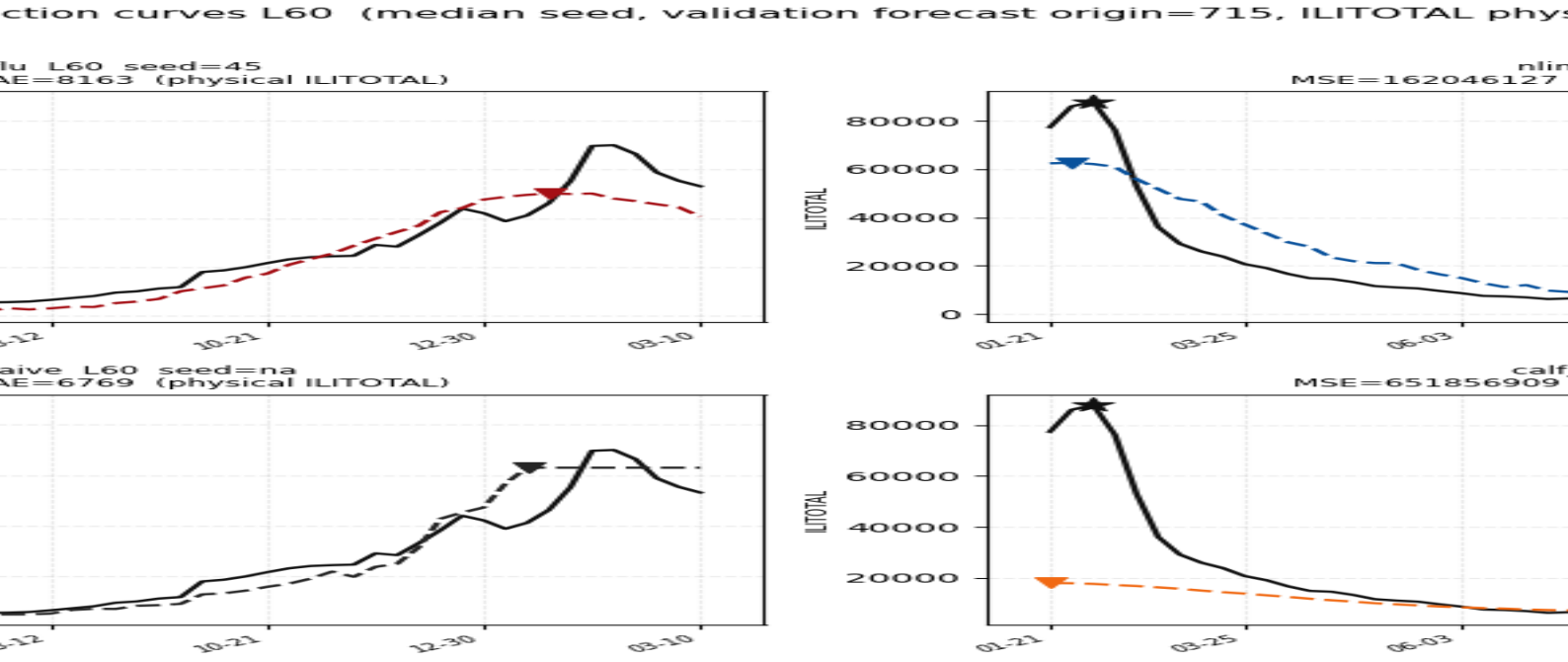


Figure 10. Prediction curves at L60 (median seed).

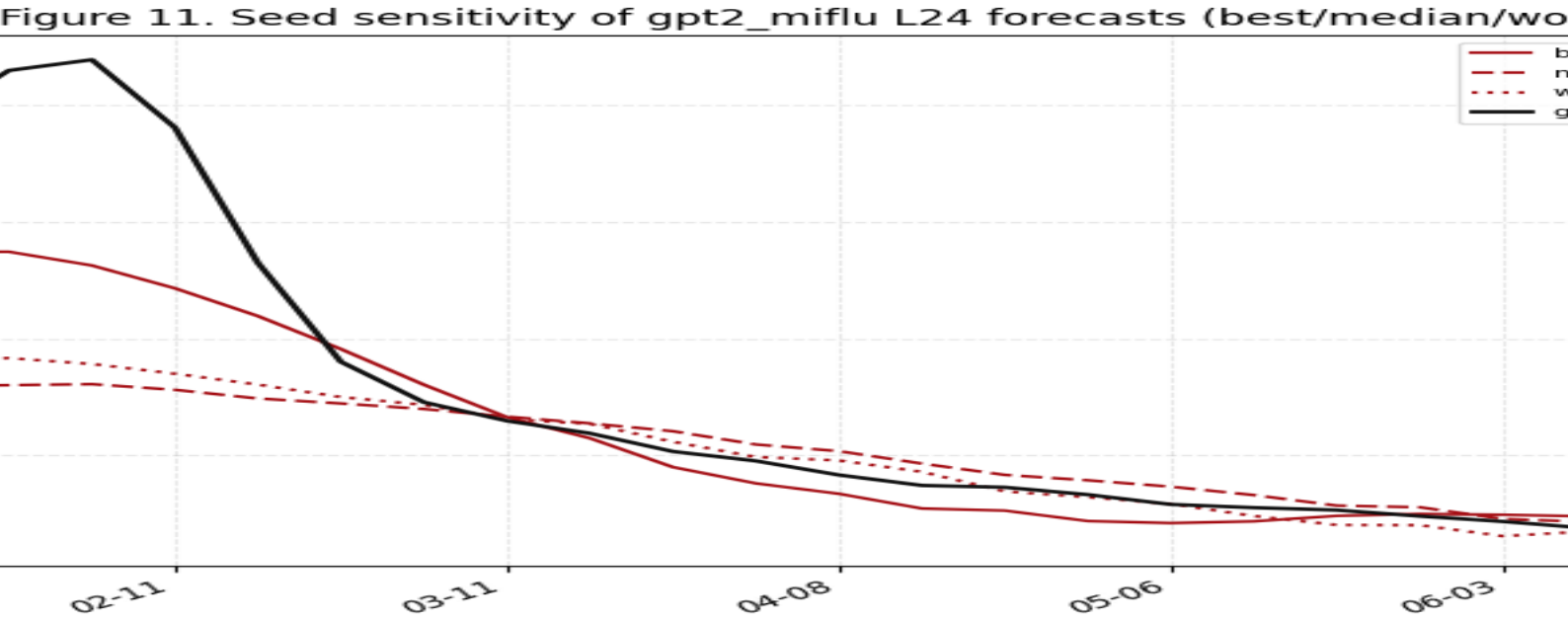


Figure 11. Seed sensitivity of gpt2_miflu L24 (best/median/worst seed).

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4. Statistical Methods

- Per-horizon point estimate = mean of 10 frozen seeds (42–51).
- `average_MSE_4L` = arithmetic mean of the four per-horizon means.
- 95% CI = seed-level t-interval (t-distribution, df = 9).
- Winner-vs-runner-up and family comparisons = matched-seed **paired t-tests** (n = 10 pairs) with Cohen's d (pooled SD).
- Ablations = matched-seed paired t vs parent; controls = matched (L, seed) comparison.
- No multiple-comparison correction was applied to the exploratory ablation/control set; the single significant ablation survives Bonferroni at $\alpha = 0.05/6 = 0.0083$.

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5. Limitations & Pending Work

1. **PTE/PIE reconstructed from durable evidence.** The durable `validation_predictions.csv` stores `target_date = None`, so peak metrics were not directly aggregatable. The target week was therefore reconstructed as `forecast_origin + 104 + step` (verified against raw ILITOTAL, inverse-transform |diff| < 1e-3) and PTE/PIE were computed on the inverse-transformed ILITOTAL channel over two complete validation seasons (2016-2017, 2017-2018). Reported as secondary application metrics only; never used in selection.
2. **18 deferred negative controls** are pending Wave-2 (parent binding = BEST_VALIDATION_LLM_AFTER_SCREENING, which is `gpt2_miflu`). They are not run here.
3. **Test/holdout are sealed** (access 0/0). No final publication champion is frozen; the Wave-1 winner is an interim screening result only.
4. Negative controls are **exploratory** (n = 6 cells, only L24/L60 × seeds 42–44); their conclusions are directional, not confirmatory.
5. `patchtst` is a "PatchTST-inspired" reimplementation (no self-supervised pretraining, no RevIN, reduced capacity) — the fidelity caveat must travel with any PatchTST claim.

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6. Conclusions

1. Under the frozen selection rule, the Wave-1 winner is `gpt2_miflu` (avg MSE 4L = 0.771), with `nlinear` as the best Non-LLM (1.070) and overall runner-up. The gap is large and highly significant ($p = 3.13\times10^{-7}$).
2. The headline should **not** be "LLMs beat classical forecasters": the Non-LLM family is significantly more accurate than the LLM family ($p = 1.44\times10^{-6}$); the win is one model, not the paradigm.
3. The gpt2_miflu advantage over a zero-parameter seasonal_naive baseline is **negative at L24/L36/L48** and positive only at L60 — seasonality explains much of the short-horizon accuracy.
4. The only text component that matters is the **dataset information** (+12.85%, p = 0.004); LoRA and multimodal fusion are statistically inert here.
5. The manuscript must report the runtime/storage trade-off (gpt2_miflu ≈ 409× nlinear parameters, 28× training time) and the seed instability of the LLM winner (SD 0.071) honestly.
6. On the secondary peak metrics, the MSE champion is not the best peak forecaster: nlinear leads on peak intensity (PIE 17.7%) and patchtst on peak timing (PTE 1.61 weeks); gpt2_miflu is mid-table (PTE 2.12 weeks, PIE 41.5%). seasonal_naive's low MSE does not transfer to peak timing (PTE 2.38 weeks).

Artifacts: ranking/summary tables and figures are under results/v3_clean/analysis/ (model_ranking.csv, primary_summary_statistics.csv, ilitotal_ranking.csv, selection_outcome.json, ablation_effects.csv, baseline_comparison.csv, resource_storage.csv, pte_pie_long.csv, pte_pie_aggregate.csv, pte_pie_audit.json, figures/, and the full 580-panel gallery under figures/gallery/). The machine-readable protocol snapshot is frozen_analysis_protocol.json.

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7. Recommended Next Steps

1. **Tutor sign-off on narrative framing** — confirm the paper claims "gpt2_miflu (a specific LLM adapter) beats baselines" rather than "LLM methods beat classical methods"; decide whether seasonal_naive becomes a formal main-table baseline.
2. **Wave-2 deferred controls** — bind the deferred parent (BEST_VALIDATION_LLM_AFTER_SCREENING = gpt2_miflu) and run the 18 deferred negative controls; do not finalize mechanism conclusions before they complete.
3. **Manuscript drafting** — draft the results section against the tables/figures here, carrying the seed-SD, seasonal-baseline, and cost-trade-off caveats verbatim.
4. **Test/holdout remains sealed** — only after the deferred controls are reconciled and the freeze audit passes should a final publication champion be considered. Do not call the Wave-1 champion the final champion.

Appendix A — Seed-level stability

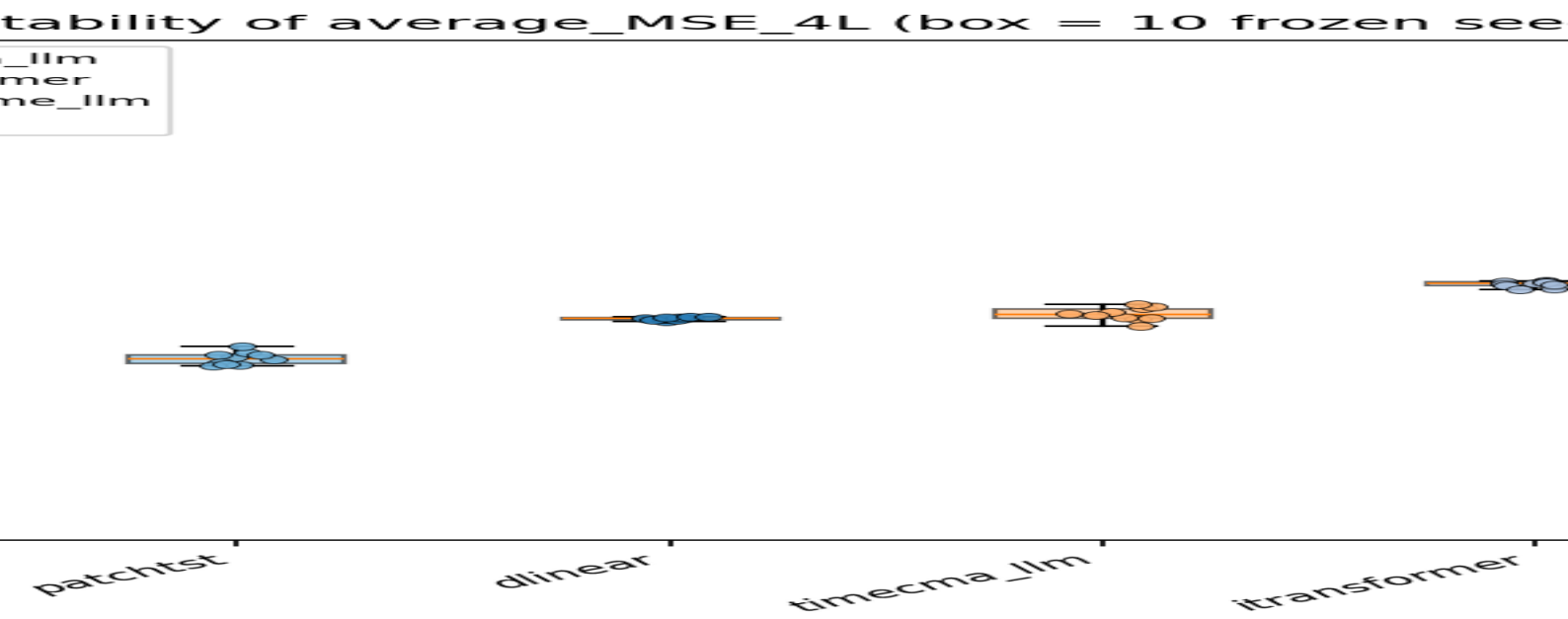


Figure 6. Seed-level stability of average_MSE_4L (box = 10 frozen seeds, dots = individual seeds). Lower MSE is better. Validation-only.

Appendix B — Accuracy vs runtime / storage

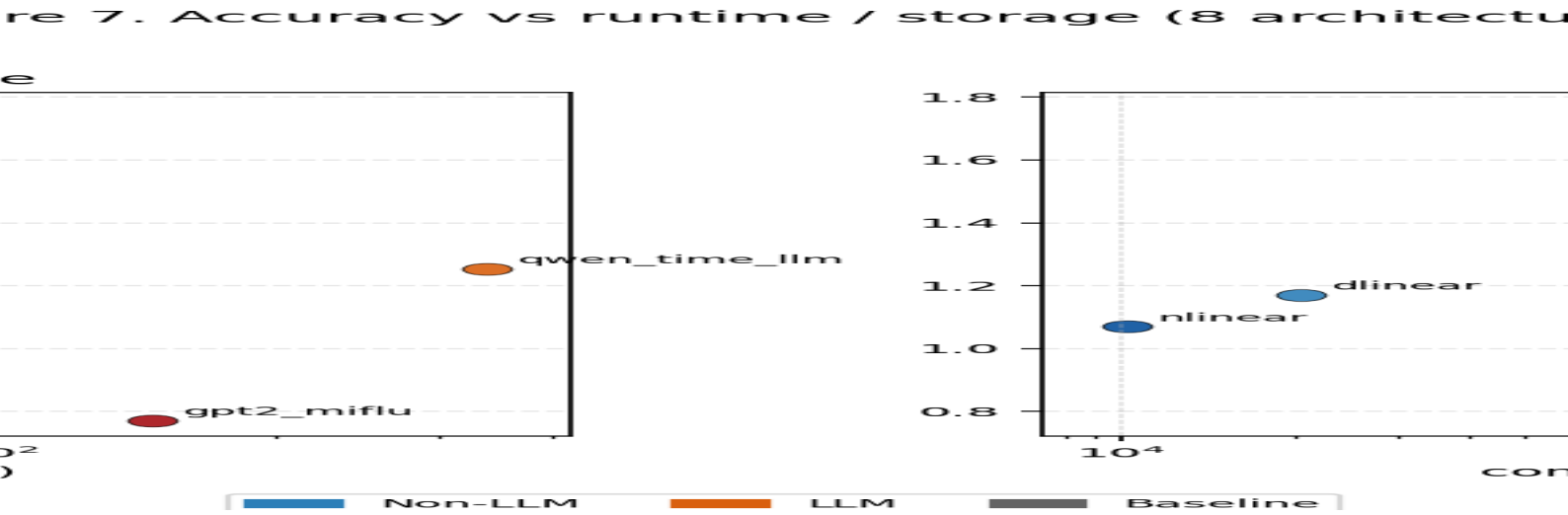


Figure 7. Accuracy vs runtime (a) and compact storage (b) for the 8 architectures. Lower MSE is better. Validation-only.

Appendix C — Wave-1 workflow and evidence-integrity

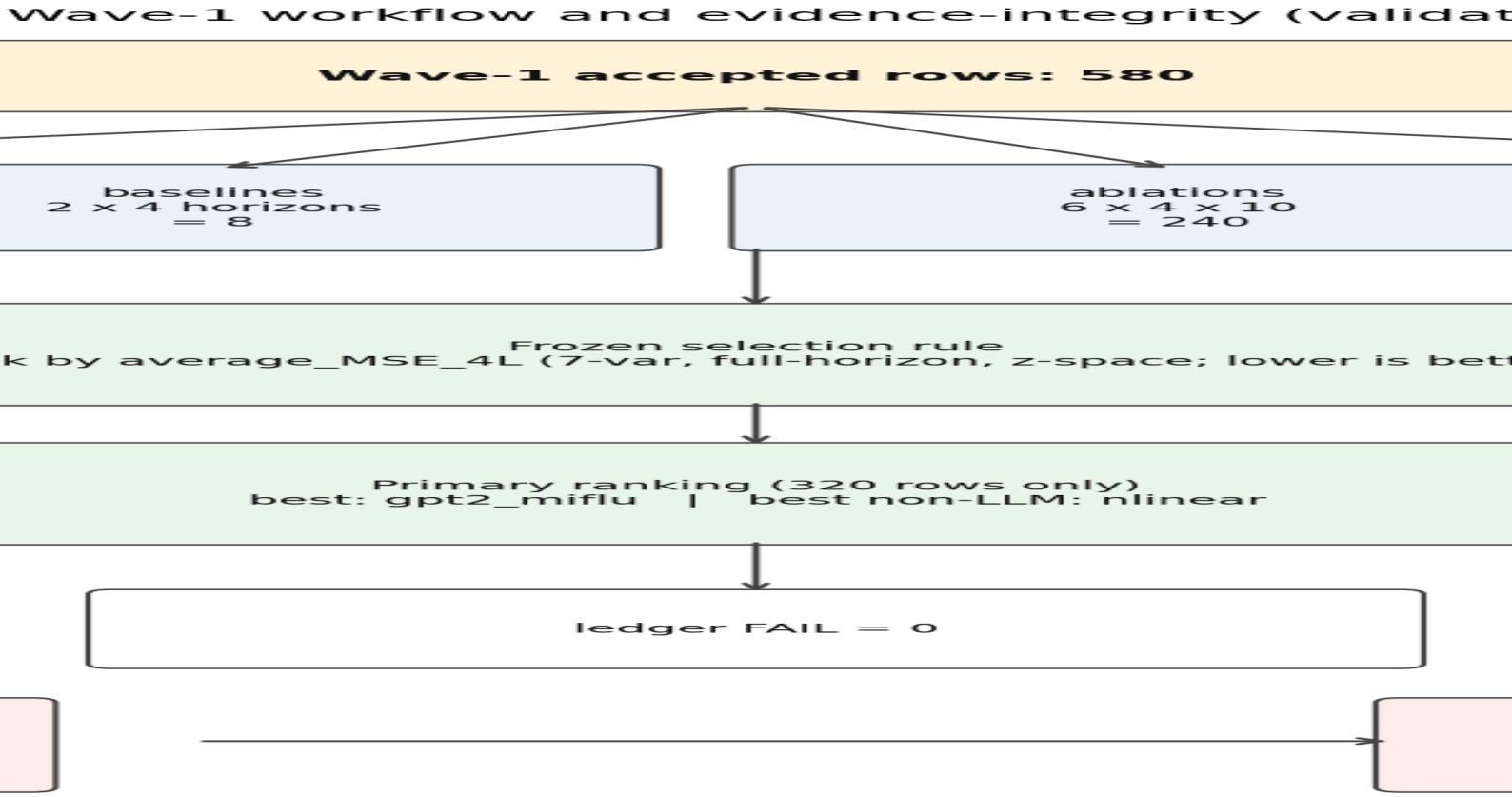


Figure 8. Wave-1 workflow and evidence-integrity. Validation-only.